

Predicting Injury Severity from Automated Driving Systems (ADS) Incidents

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Abstract

This study develops a random forest to predict injury severity levels (**Minor, Moderate, Severe**) in **ADS** incidents data from **1,489 NHTSA reports**. Including both structured (weather, speed, motion) and unstructured data (crash narrative), our model identifies key predictive factors for each incident.

Introduction

- Compulsory crash reporting to NHTSA
- Imbalance to be addressed (90.4% **Minor**, 2.8% **Moderate**, 5.8% **High**)
- **LDA** (Latent Dirichlet Allocation) to understand **probability of topics** within each type of accident and their **differences**
- Interaction features created to capture **combined effects**

Methodology

We use of **SMOTE** to address class imbalance. Then implemented feature engineering and **one-hot encoding** to structured data fields (road type, collision object, etc.) and **preprocess narrative** text. Also, we removed **multicollinear** features to real-world accuracy

Ethical Considerations

Crash reporting data needs **Complete disclosure and detail of the incident**

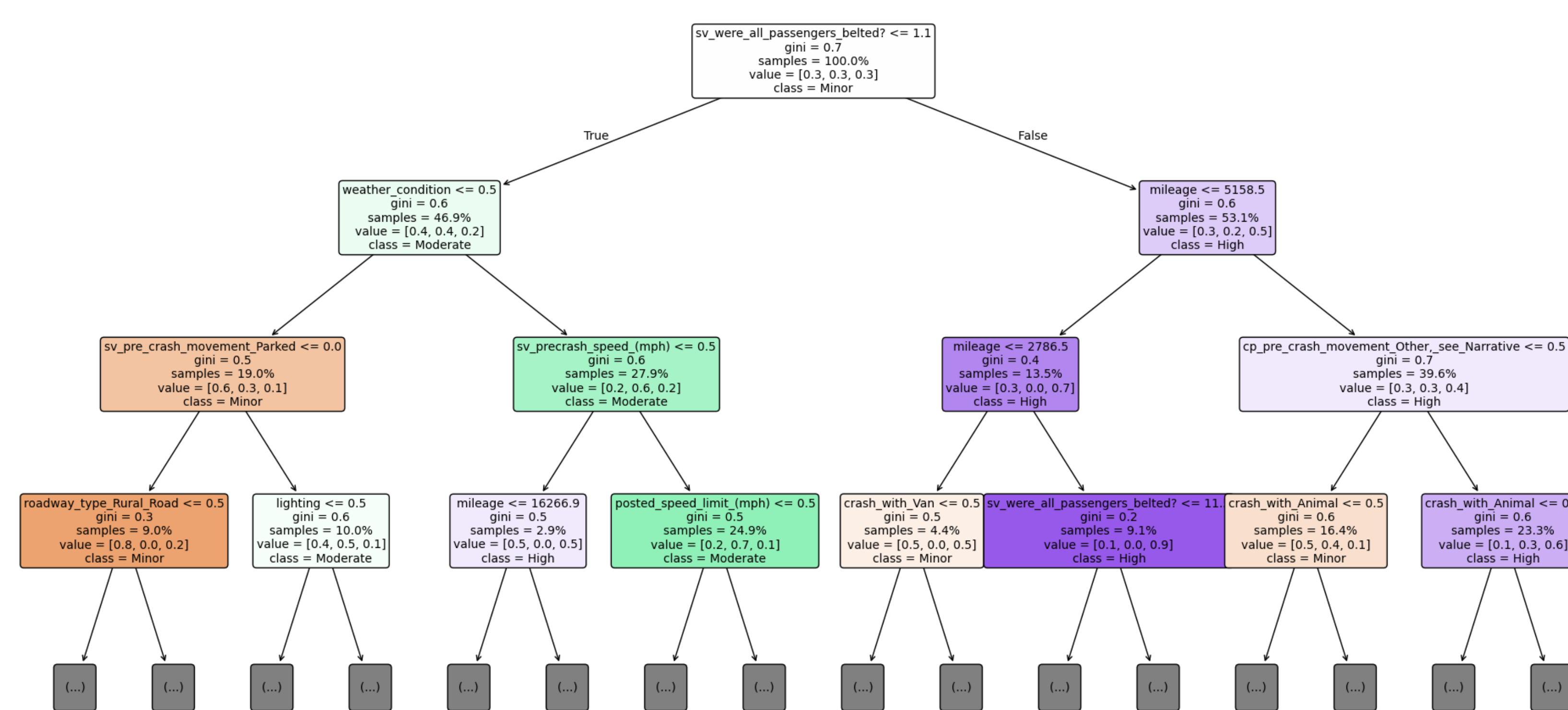
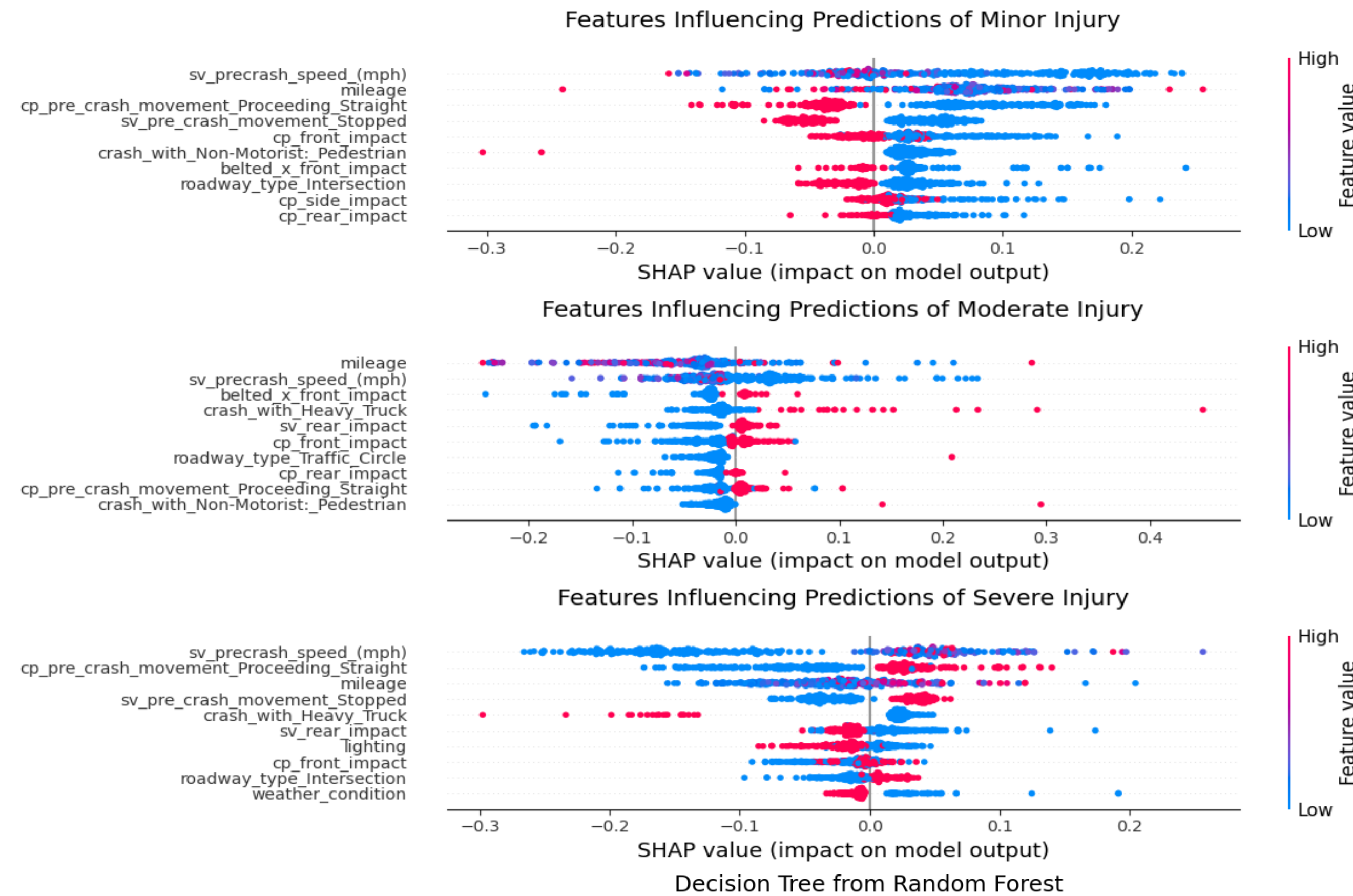


Figure 1 — Feature Importance

SHAP Values that gives how much a feature has predicting power and in which direction. For severe, we see front impacts have high prediction power. For moderate, pre-crash speed has prediction power.

Figure 2 — Decision Tree

Based on Gini impurity, the RF has an ensemble of trees that make splits differently. In this scenario the presence of seatbelt determines how the prediction follows a path to Minor, Moderate or Severe injuries.

Results

- **Overall accuracy 93%** (96% Minor, 69% Moderate, 50% High)
- **Seatbelt** and Impact location (**sideways**) emerged as strong predictor.
- **Pre-crash speed** and specific **pre-crash movements** were highly predictive. Involvement of **heavy trucks -> Severe**

Limitations

- Critical class **imbalance** towards minor injury type of incidents
- **Redacted** Data hinder deeper research for specific companies
- Nuanced factors (speed – > injury) need further review to understand how to interpret them

Conclusion

- There is a need for **further analysis** with qualitative data analysis.
- As well as a deeper **understanding of the narratives** with topic modelling (casts a light on how features interact speed->injury)
- **Redaction** from manufacturers, can cast shadow on revealing datasets
- They only way to improve safety is through **disclosing information** and analyzing it

References

References

- [1] National Highway Traffic Safety Administration. General Order on Crash Reporting. <https://www.nhtsa.gov/laws-regulations/standing-general-order-crash-reporting>
- [2] Alambeigi et al. (2020). *Crash themes in Automated Vehicles*. Available online: <https://arxiv.org/abs/2001.11087>